Research Protocol v1.0

Design and Synthetic Evaluation of a Clinician-Supervised EHR/FHIR Workflow for Early Pediatric Vision Risk Support

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Registered baseline: GitHub pre-release v0.1, commit 0c30d7c
https://github.com/nipa-analytics/pediatric-vision-screening-ai/releases/tag/v0.1

1. Protocol synopsis

Design	**In-silico, controlled scenario-based feasibility and safety evaluation; no patient enrollment.**
Population	Synthetic pediatric cases ages 0–5 years, with predefined under-3 subgroup.
Clinical scope	Workflow support for strabismus and amblyopia-related concerns, not diagnosis.
Architecture	HL7 FHIR R4 resources and structured audit events, as supported by implementation.
Human oversight	Clinician review and final referral disposition retained; no autonomous clinical action.
Primary outcomes	Scenario-level workflow correctness, appropriate abstention, explanation/traceability completeness.
Secondary outcomes	FHIR conformance, audit completeness, referral-state integrity, subgroup consistency.
Clinical validation	Out of scope; no claims of diagnostic accuracy, effectiveness, or real-world safety.

2. Rationale and research question

Early pediatric vision concerns require age-appropriate screening, escalation pathways, and careful interpretation of incomplete clinical information. The present prototype is a technical research workflow; its proposed synthetic evaluation tests behavior against predetermined requirements rather than estimating disease risk or clinical benefit.

Research question: Can a clinician-supervised EHR/FHIR pediatric vision workflow provide transparent risk support, appropriately abstain when information is insufficient, and preserve clinician authority across controlled synthetic scenarios?

3. Objectives and hypotheses

Primary objective: Quantify whether the workflow meets prespecified safety and process expectations over controlled synthetic scenarios.

- H1: For scenarios with a defined expected workflow state, output and transitions will match the reference specification.
- H2: Scenarios designed to have insufficient or conflicting information will trigger the predefined abstention or escalation response rather than unsupported certainty.
- H3: Every evaluated recommendation or abstention event will be traceable to input fields, rule/version, and clinician-review status.

4. Study population and scenario construction

Unit of analysis: one synthetic pediatric encounter/scenario. Proposed initial sample: 150 deterministic scenarios (40 routine, 30 elevated-concern, 30 missing-data, 25 conflicting-data, 25 targeted subgroup/stress scenarios). These are engineering coverage targets rather than a statistically powered clinical sample; overlapping subgroup characteristics will also be examined across all cases.

Inclusion: synthetic age 0–5, valid scenario identifier, documented expected workflow outcome, and inputs conforming to the scenario schema. Exclusion: real patient data, undocumented expected outcomes, and scenarios requiring unsupported clinical judgments. Malformed FHIR inputs are retained in a separate negative-testing set, not silently excluded.

Prespecified stratification: younger than 3 versus 3–5 years; completeness; conflicting information; scenario concern tier; where synthetically modeled and ethically appropriate, language/accessibility or other workflow-relevant dimensions. No prevalence estimates or demographic fairness claims will be inferred from synthetic counts.

5. Intervention and comparator

Intervention: tagged prototype v0.1, executed with documented configuration and deterministic test fixtures. Any proposed changes to behavior are evaluated separately as v0.2, never retroactively attributed to v0.1.

Comparator/reference standard: independently written expected workflow outcomes and decision tables approved before execution, based on technical requirements and evidence review where applicable. These represent engineering specifications, not ophthalmologist-confirmed diagnoses.

6. Endpoint definitions

Measure	Operational definition
Workflow correctness	Number of scenarios matching all applicable expected states / number of evaluable scenarios.
Abstention appropriateness	For required-abstention cases: true abstentions / all required-abstention cases. Also report inappropriate abstention among cases where a supported response is expected.
Traceability completeness	Evaluated events with complete recorded input provenance, reason/rule version, output, timestamp and review status / all evaluated events.
FHIR conformance	Resources passing FHIR R4 structural, cardinality, terminology and relevant profile validation / resources tested; report rule categories and failures.
Referral-state integrity	Allowed referral-state transitions / all observed referral-state transitions; count unauthorized transitions separately.
Audit completeness	Expected auditable actions containing defined audit fields / all expected auditable actions.
Subgroup consistency	Descriptive endpoint rates by prespecified strata, with denominators and uncertainty; flag differences for review, not fairness certification.

7. Safety controls and failure handling

- No autonomous clinical diagnosis, treatment, or final referral authorization.
- Hard-stop or abstention on prespecified missing, contradictory, or invalid critical inputs; document the exact condition and expected response.
- Require human confirmation for referral disposition and retain the original recommendation plus overrides.
- Use synthetic data only; prohibit accidental inclusion of patient identifiers or credentials in fixtures/logs.
- Capture software version, timestamp, relevant input provenance, exception handling and reproducible test result.

8. Data model and quality assurance

Document each variable in a companion Evidence Matrix/Data Dictionary: name, definition, type, allowed values, missingness handling, age relevance, FHIR mapping, supporting source, evidence quality, and inclusion decision. Candidate variables (not validated weights) include family history, prematurity, observed alignment concern, prior screening concern, and age at assessment.

Validation should use FHIR R4 terminology, cardinality and structural rules; preserve validation output for each case. Capture code version, fixture seed (if randomized), environment, dependency versions, and run date. Maintain independent expected-outcome fixtures to reduce confirmation bias.

9. Analysis plan and decision criteria

Present counts, numerators/denominators, percentages, and scenario-level failure tables for every endpoint. Provide exploratory binomial confidence intervals when meaningful but do not interpret them as population-level clinical estimates. Review all safety-critical failures individually. Report under-3 metrics separately.

Provisional engineering gates (to be approved before study execution): zero unauthorized autonomous referrals; zero silent unsupported outputs in required-abstention fixtures; 100% provenance for safety-critical actions; all FHIR validation failures disclosed and classified. Other performance thresholds should be set prospectively after review of the baseline implementation rather than selected post hoc.

10. Ethics, governance, and limitations

This proposed study uses wholly synthetic data and involves no recruited participants, patient records, or deployment in clinical care. An institutional determination regarding human-subjects oversight should be sought if required by the host institution; do not self-certify exemption for future real-world work.

Limitations include scenario designer bias, incomplete simulation of pediatric clinical complexity, absence of clinician/patient usability testing, lack of estimates of diagnostic performance, lack of clinical outcomes, and inability to establish real-world subgroup equity or generalizability.

11. Reproducibility and change control

Baseline: GitHub tag v0.1 (commit 0c30d7c). Preserve release artifacts and fixed fixture/test manifest. Any modification to rules, datasets, or endpoints requires a dated protocol amendment and a new version; never move the v0.1 tag. Report failures and deviations transparently.

12. Deliverables and sequence

1. Protocol v1.0 and companion evidence matrix/data dictionary.
2. Scenario specification and reference expected-outcome manifest (target 150 initial scenarios).
3. Automated test suite with FHIR validation, safety/abstention checks, audit checks and versioned logs.
4. Results tables, limitation statement and v0.2 change proposal.
5. Methods/prototype manuscript after evaluations are completed.

13. Reporting frameworks and source references

- Vasey B, et al. DECIDE-AI: reporting guideline for early-stage live clinical evaluation of AI decision support systems. BMJ. 2022;377:e070904. https://www.bmj.com/content/377/bmj-2022-070904 (principles only; this study is not a live clinical evaluation).
- HL7. FHIR R4 v4.0.1 — Validation. https://hl7.org/fhir/R4/validation.html
- Collins GS, et al. TRIPOD+AI statement. BMJ. 2024;385:e078378. https://www.bmj.com/content/385/bmj-2023-078378 (for future prediction-model studies; not applicable to this workflow-only evaluation).

Protocol note: This document proposes an evaluation; no study execution, outcomes, external clinical review, or regulatory authorization is claimed.